Two-Branch Café Sales Insights

A café brand operates two sales streams:

1. Stream A (index_1)
2. Stream B (index_2)

Management wants a report that –

1. Compares both the streams
2. Understands sales, products and payment mix over time (But not limited to)
3. Run simulations to plan promotions

Focus on:

1. Create a clean Date table from min/max across both files (supports daily, MTD/QTD/YTD, YoY). Parse date/datetime into a proper Date and Time (derive Hour if needed).
2. Create a unified Fact table by unioning the two sources with a new column –
   a. source_stream = "Stream A" for index_1.csv, "Stream B" for index_2.csv.
   b. Keep card column; it will be blank for Stream B.
3. **Relationships**:

Date dimension ↔ Fact on Date.

Product dimension ↔ Fact on coffee_name.

Payment dimension ↔ Fact on cash_type.

Goal: a star schema with one FactSales and small dimensions.

4. **Implement measures for sales performance using the Date table:**

- Total Sales (from `money`).
- MTD / QTD / YTD Sales (current period).
- Same period last year**:** YoY Sales**,** YoY % for MTD/YTD.
- Rolling windows**:** last 7 days and 28 days total sales.
- Daypart: derive Morning/Afternoon/Evening/Night from parsed time, then compare sales by daypart over time**.**

Goal: All measures must respect slicers (date, product, payment, source).

5. **Sync slicers: Create three report pages**

- Overview (global KPIs & time windows),

- Product Focus (by `coffee_name`),
- Payments/Customer Focus (by `cash_type`, and card logic for Stream A).

Sync these slicers across pages:

- **Date** (from Date table),
- **Source Stream** (source_stream),
- **Coffee Name**,
- **Payment Type** (cash_type).

Goal: Changing a slicer on one page instantly reflects on the other pages.

6. Create four bookmark scenarios that only use data in these files:

MTD vs YTD View Toggle

State A highlights MTD metrics; State B highlights YTD metrics.

Bookmarks must switch between the two states on the same page without duplicating pages.

Show/Hide Filters

One bookmark shows all slicers (analysis mode).

Another hides slicers (presentation mode).

Top Products vs Payment Mix

State A focuses on product performance (by coffee_name).

State B focuses on payment split (by cash_type).

Must work with the same slicer state.

Reset to Default

A bookmark that clears selections back to the agreed default (e.g., All dates, All streams).

Acceptance: Bookmarks must be set with appropriate options (data/display) so they **don't accidentally overwrite slicer sync** unless intended.

7. **Parameters:**
   A) Field Parameters (dimension/measure switching)
   Create at least two field parameter experiences:

1. **Breakdown Switcher**: toggle the analysis axis between coffee_name, cash_type, and source_stream.
2. **Metric Switcher**: toggle the metric between Total Sales, YoY %, MTD Sales, Rolling 28-day Sales.

   Goal:

   A) single visual can switch its axis or measure based on the parameter selection.

   B) What-If Parameter (simple simulation—must use only available data)

   Create a **Price Uplift %** parameter (e.g., 0–20%).

- Apply it to compute Simulated Sales = money adjusted by uplift (only affects shown totals; it doesn't change the data).
- Provide Delta between Simulated Sales and Actual Sales.
- Ensure it respects all slicers and both streams.

  No external cost/quantity fields—this is a pure revenue simulation compliant with the files.

Must Solve:

• **Parsing**: Normalize `date`/`datetime` to robust Date & Time; handle rows where `datetime` is time-only.

• **Types**: Ensure `money` is **numeric/decimal** after union (Stream B is int; Stream A is float).

• **Card Nulls**: Keep blanks for Stream B; avoid relationship errors.

• **Filters**: DAX should use safe division and respect context.

• **Calendar**: Date table must span min→max across both streams and be marked as **Date table**.

Deliverables:

1. Visuals of choice
2. **Data model** (Fact + Date + Product + Payment + optional Card).
3. **DAX measures** implementing time intelligence, rolling windows, YoY, and simulation.
4. **Slicer sync** working across three pages.
5. **Two parameter experiences** (Field & What-If) fully wired.
6. **Short README** (bulleted) explaining:
   - **a.** model decisions,
   - **b.** assumptions for time parsing/dayparts,
   - **c.** how bookmarks & parameters are intended to be used.